

Machine Learning Practical Test Paper

Duration: 2 Hours

Maximum Marks:100

Section A – Python & Data Handling (20 Marks)

Q1. Create a Pandas DataFrame using the given data, display first 5 rows, check missing values, dataset info, summary statistics.

Q2. Separate X and y, split 80:20, apply StandardScaler.

Section B – Machine Learning Models (50 Marks)

Q3 Linear Regression: Train, predict, MAE, MSE, RMSE, R2.

Q4 Logistic Regression: Train, predict, Accuracy, Confusion Matrix.

Q5 KNN: Train, predict, Accuracy for k=3 and k=5.

Q6 Decision Tree: Train, predict, Accuracy, Display Tree.

Q7 Random Forest: Train, predict, Accuracy, compare with Decision Tree.

Section C – Visualization (15 Marks): Histogram, Box Plot, Scatter Plot, Heatmap, Count Plot.

Section D – Evaluation (15 Marks): Accuracy, Precision, Recall, F1, Confusion Matrix, Classification Report. Answer: Best model, why, overfitting, underfitting, improvement.

Bonus (5): Train SVM and compare with Logistic Regression and KNN.